Algebra II with Trigonometry Chapter 6.2

Olympiad Solutions (Gemini)

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Solutions

1. Suppose t is a real number such that $\sec t + \tan t = 2024$. Evaluate the exact value of $\sec(-t) + \tan(-t)$.

Solution: Recall the even-odd parity properties of the secant and tangent functions: $\sec(-t) = \sec t$ and $\tan(-t) = -\tan t$. We are tasked with evaluating $\sec t - \tan t$. By the fundamental Pythagorean identity, we know that $\sec^2 t - \tan^2 t = 1$. Factoring this difference of squares yields:

$$(\sec t - \tan t)(\sec t + \tan t) = 1$$

Substituting the given condition $\sec t + \tan t = 2024$, we obtain:

$$2024(\sec t - \tan t) = 1 \implies \sec t - \tan t = \frac{1}{2024}$$

Thus, the exact value of $\sec(-t) + \tan(-t)$ is $\frac{1}{2024}$.

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2. Let $f(x) = \frac{x^3 \sec x}{1+\cos^2 x} - 5\csc(x)\sin^2(x) + 10$. Evaluate the exact sum of $f\left(-\frac{\pi}{5}\right) + f\left(\frac{\pi}{5}\right)$.

Solution: First, simplify the second term of the function: $5\csc(x)\sin^2(x) = 5\left(\frac{1}{\sin x}\right)\sin^2(x) = 5\sin x$. Thus, we can rewrite $f(x)$ as $f(x) = g(x) + 10$, where

$$g(x) = \frac{x^3 \sec x}{1+\cos^2 x} - 5\sin x$$

Next, let us examine the parity of $g(x)$. The function x^3 is odd, $\sec x$ is even, and $1+\cos^2 x$ is even, making the first term $\frac{x^3 \sec x}{1+\cos^2 x}$ an odd function. The second term $5\sin x$ is also odd. Since the sum of odd functions is odd, $g(x)$ is an odd function, satisfying $g(-x) = -g(x)$. We want to evaluate:

$$f\left(-\frac{\pi}{5}\right) + f\left(\frac{\pi}{5}\right) = \left[g\left(-\frac{\pi}{5}\right) + 10\right] + \left[g\left(\frac{\pi}{5}\right) + 10\right]$$

Since $g\left(-\frac{\pi}{5}\right) + g\left(\frac{\pi}{5}\right) = 0$, the sum evaluates exactly to:

$$10 + 10 = 20$$

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3. Evaluate the exact value of the following nested composition:

$$\tan\left(\frac{\pi}{4}\sec\left(\frac{\pi}{3}\cos\left(\pi\sin\left(\frac{7\pi}{6}\right)\right)\right)\right)$$

Solution: We evaluate the nested expression elegantly from the inside out: 1. Evaluate the sine: $\sin\left(\frac{7\pi}{6}\right) = -\frac{1}{2}$ 2. Multiply by π: $\pi\left(-\frac{1}{2}\right) = -\frac{\pi}{2}$ 3. Apply cosine: $\cos\left(-\frac{\pi}{2}\right) = 0$ 4. Multiply by $\frac{\pi}{3}$: $\frac{\pi}{3}(0) = 0$ 5. Apply secant: $\sec(0) = 1$ 6. Multiply by $\frac{\pi}{4}$: $\frac{\pi}{4}(1) = \frac{\pi}{4}$ 7. Apply tangent: $\tan\left(\frac{\pi}{4}\right) = 1$

The exact value of the composition is 1.

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4. The real number t satisfies the inequality $\tan t + \cot t < 0$. If it is also known that $\sec t > \csc t$, determine the quadrant in which the terminal point of t lies.

Solution: Rewrite the first inequality in terms of sine and cosine:

$$\tan t + \cot t = \frac{\sin t}{\cos t} + \frac{\cos t}{\sin t} = \frac{\sin^2 t + \cos^2 t}{\sin t \cos t} = \frac{1}{\sin t \cos t}$$

For $\tan t + \cot t < 0$, we must have $\sin t \cos t < 0$. This implies that $\sin t$ and $\cos t$ have opposite signs, which strictly restricts the terminal point of t to either Quadrant II or Quadrant IV.

Next, analyze the second condition: $\sec t > \csc t \implies \frac{1}{\cos t} > \frac{1}{\sin t}$.

- If t is in Quadrant II, $\cos t < 0$ and $\sin t > 0$. This gives $\sec t < 0$ and $\csc t > 0$, yielding $\sec t < \csc t$, which contradicts the given inequality.
- If t is in Quadrant IV, $\cos t > 0$ and $\sin t < 0$. This gives $\sec t > 0$ and $\csc t < 0$, yielding $\sec t > \csc t$, which safely satisfies the condition.

Therefore, the terminal point of t lies in Quadrant IV.

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5. For the real number $t = \frac{2024\pi}{13}$, determine the exact value of the expression:

$$\sec^4 t - \tan^4 t - 2\tan^2 t$$

Solution: We can simplify the given algebraic expression by recognizing the difference of squares in the first two terms:

$$\sec^4 t - \tan^4 t = (\sec^2 t - \tan^2 t)(\sec^2 t + \tan^2 t)$$

Using the identity $\sec^2 t - \tan^2 t = 1$, this effortlessly reduces to:

$$1 \cdot (\sec^2 t + \tan^2 t) = \sec^2 t + \tan^2 t$$

Substitute this back into the full expression:

$$(\sec^2 t + \tan^2 t) - 2\tan^2 t = \sec^2 t - \tan^2 t = 1$$

The expression evaluates invariantly to 1 for all values of t where $\sec t$ and $\tan t$ are defined. Since the denominator 13 is odd, $t = \frac{2024\pi}{13}$ is not an odd multiple of $\frac{\pi}{2}$, meaning the functions are perfectly well-defined.

The exact value of the expression is 1.

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6. A real number t satisfies $\sin t + \cos t = \frac{1}{3}$. Determine the exact value of $\tan t + \cot t$.

Solution: First, rewrite the target expression in terms of sine and cosine:

$$\tan t + \cot t = \frac{\sin t}{\cos t} + \frac{\cos t}{\sin t} = \frac{\sin^2 t + \cos^2 t}{\sin t \cos t} = \frac{1}{\sin t \cos t}$$

To find the required value of $\sin t \cos t$, we simply square both sides of the given equation $\sin t + \cos t = \frac{1}{3}$:

$$(\sin t + \cos t)^2 = \left(\frac{1}{3}\right)^2$$

$$\sin^2 t + 2\sin t \cos t + \cos^2 t = \frac{1}{9}$$

Applying the identity $\sin^2 t + \cos^2 t = 1$, we have:

$$1 + 2\sin t \cos t = \frac{1}{9}$$

$$2\sin t \cos t = \frac{1}{9} - 1 = -\frac{8}{9}$$

$$\sin t \cos t = -\frac{4}{9}$$

Taking the reciprocal, we find the exact value:

$$\tan t + \cot t = \frac{1}{-4/9} = -\frac{9}{4}$$